Hazem Masarani

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Research

PACE Lab, Stony Brook University Oct 2025 – Present
Graduate Research Fellow Stony Brook, NY

- Implemented Mamba and Mamba-2 tensor-parallel inference with PyTorch Distributed and NCCL.
- Benchmarked latency, throughput, and correctness on NVIDIA A6000 and A100 GPUs.
- Implemented SSM–Gate parallelism using CUDA and Triton; preliminary benchmarks showed gains over conventional tensor parallelism at small batch sizes.

Experience

Parasail Sep 2026 – Present
Technical Writer, LLM Inference (Freelance) Remote

- Write about inference performance, throughput–latency trade-offs, and speculative decoding.

Outlier Aug 2024 – Present
Prompt Evaluation and Response Optimization (Freelance) New York, United States

- Evaluate AI-generated responses and document quality feedback for RLHF projects.

Al-Madinah School Queens Sep 2024 – Jun 2025
Computer Teacher – Elementary School Queens, NY

- Taught computing fundamentals through practical lessons adapted to students' learning needs.

e& (Etisalat) Feb 2024 – Sep 2024
Senior Automation & Network Data Engineer Egypt (Hybrid)

- Built Python pipelines and monitoring tools to analyze network alarms, topology, and outages.

e& (Etisalat) Jul 2023 – Jan 2024
Junior Automation & Data Engineer Egypt (Hybrid)

- Developed network-monitoring automation and maintained SQL data-processing pipelines.

Publications

Tensor Parallelism for State-Space Models — IEEE ICDCS 2026.
Anurag Dutt, Nimit Shah, **Hazem Masarani**, and Anshul Gandhi.

Projects

vLLM Mamba Extension GitHub

- Applied the SSM–Gate split in vLLM's CUDA and Triton code for Mamba inference.

SSM–Gate Parallel Inference GitHub

- Split Mamba's SSM and gate paths into separate submodels, applying tensor parallelism within each using PyTorch and NCCL.

Education

Stony Brook University May 2027 (expected)
M.S. in Data Science; GPA: 3.96/4.00 Stony Brook, NY

Alexandria University Jun 2022
B.Sc. in Computer & Communication Engineering; GPA: 3.47/4.00 Alexandria, Egypt

Skills

Programming: Python, C/C++, CUDA, Triton
Frameworks & Parallelism: PyTorch, vLLM, PyTorch Distributed, NCCL, Tensor/Model Parallelism
Tools: Linux, Git, Docker, Slurm